Problem 0.1. Let $\mathcal{M}_g$ be the moduli space of genus g Riemann surfaces. We consider the Deligne-Mumford compactification $\overline{\mathcal{M}}_g = \mathcal{M}_g \sqcup \partial\mathcal{M}_g$. The boundary $\partial\mathcal{M}_g$ admits a partition into strata Δ_G parametrizing stable curves with dual graph G.

Let $V_0 \subset \mathcal{M}_g$ be a (complex) smooth subvariety of (complex) dimension 2 that is totally geodesic with respect to the Teichmüller metric. Let V_t be a holomorphic family of subvarieties of $\mathcal{M}_g$ deforming V_0, for t in the unit disc $\Delta \subset \mathbb{C}$ [1]. We impose several technical assumptions on this family, namely (all closures below, e.g. $\overline{V}_t$, are taken in $\overline{\mathcal{M}}_g$):

(i) for every boundary stratum Δ_G, we have that $\overline{V}_t \cap \Delta_G$ is a (smoothly trivial) family, and

(ii) the families so produced are compatible with V_0 in the sense that if the boundary of V_0 in a smaller stratum arises from its boundary in a larger stratum, then the same holds for V_t. That is, if $\Delta_G, \Delta_{G'}$ are strata with $\Delta_{G'}$ on the boundary of Δ_G, and U is an open set in $\overline{\mathcal{M}}_g$ with

$$\overline{\overline{V}_0 \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V}_0 \cap \Delta_{G'} \cap U,$$

then, for all t,

$$\overline{\overline{V}_t \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V}_t \cap \Delta_{G'} \cap U.$$

Note that we do *not* assume that V_t is totally geodesic for $t \neq 0$.

Now fix Γ a dual graph; we will assume for ease of notation that Δ_Γ is a product of lower complexity moduli spaces $\mathcal{M}_{g_1,n_1} \times \cdots \times \mathcal{M}_{g_k,n_k}$ [2]. For each i, let $\pi_i : \Delta_\Gamma \to \mathcal{M}_{g_i,n_i}$ be the natural projection map.

Denote by ∂V_t the closure of V_t in $\overline{\mathcal{M}}_g$ intersected with Δ_Γ. Suppose that $\dim \partial V_t = 1$ and that each ∂V_t is constant on some factor $\mathcal{M}_{g_i,n_i}$ of Δ_Γ, i.e. for some i, $\pi_i(\partial V_t) = \{Y_t\}$ for all t. Prove that the family ∂V_t is non-varying on that factor, i.e. $\pi_i(\partial V_t) = \{Y_0\}$ for all t.

[1] i.e. a complex submanifold $\mathcal{V} \subset \mathcal{M}_g \times \Delta$, with $V_t = \psi(\phi^{-1}(t))$, where $\phi : \mathcal{V} \to \Delta, \psi : \mathcal{V} \to \mathcal{M}_g$ are the natural projections; we will further assume that this gives a smoothly trivial family of submanifolds.

[2] for some boundary strata, we would need to quotient this by a finite group; we exclude those here